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ON THE TURÁN PROBLEM FOR k-CHORDED CYCLES

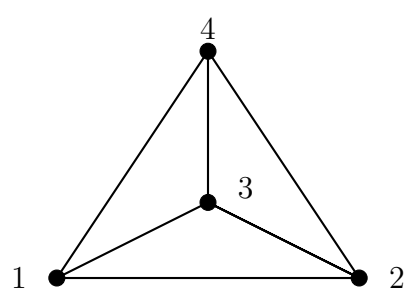
GAUTHAM ANNE, VAIBHAV MISHRA
MENTOR: BRADLEY ELLIOTT

Abstract. We examine the Turán Number of a k-chorded cycle, and provide explicit formulas for k from 0 to 6. The formulas are $n-1$, $2n-4$, $2n-3$, $2n-2$, $3n-9$, and $3n-7$, as k varies from 0 to 6, respectively. We develop several methods, such as graph decomposition, longest path theorem, and the use of extremal bipartite graphs. These methods were used by the authors frequently throughout the paper to prove the theorems for different values of k. We introduce a concept known as Keystone Turán Numbers. Finally, we present a conjecture in the form of a recursive formula for the generalized Turán Number of a k-chorded cycle.

1. Introduction

1.1. Definitions.

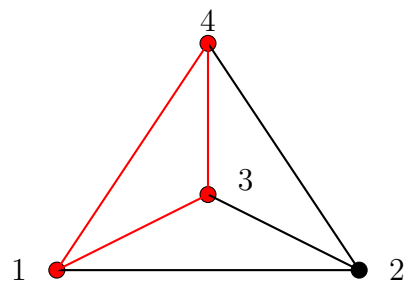
Definition 1.1 (Graph)**.** A set of vertices and edges. The image below is an example of a graph.



Note that in this paper a any referral to a graph means a finite, undirected graph with no loops contained inside it. We also do not consider graphs with double edges, such as the one below:

Definition 1.2 (Subgraph)**.** A graph with all edges and vertices that are also part of a larger graph. The red graph is a subgraph of the overall graph.

Date: June 20, 2021.



Note that if a subgraph contains a set number of chords then the graph also contains at least that amount of chords. So if we prove that a subgraph has a certain amount of chords it is sufficient to prove that the graph in general has that amount of chords too.

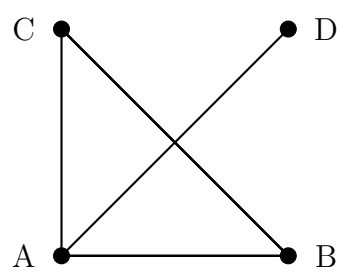
Definition 1.3 (Path)**.** A path is a sequence of vertices with the property that each vertex in the sequence is adjacent to the vertex next to it. Below is an example of a path:

Definition 1.4 (Degree)**.** Let the degree of a vertex v_i (denoted as $\deg(v_i)$) equal the number of edges connected to v_i.

Definition 1.5 (Degree Sum)**.** The degree sum of a graph is the sum of the degrees of all vertices.

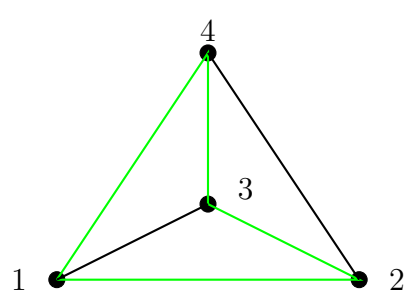
Definition 1.6 (Minimum Degree)**.** The minimum degree is the degree of the vertex with the least number of edges incident to it (denoted as δ).

Definition 1.7 (Leaf)**.** Any vertex with a degree equal to 1. Vertex D has a degree of 1, and is considered a leaf.



Definition 1.8 (Cycle)**.** A path that starts and ends with the same vertex.

Definition 1.9 (Hamiltonian Cycle)**.** A cycle that goes through each vertex exactly once. A Hamiltonian cycle is highlighted below in green:



Definition 1.10 (Chord)**.** An edge between whose vertices that define it lie on a cycle.

Definition 1.11 (k-Chorded Cycle)**.** A cycle with k number of chords inside it.

Definition 1.12 (Block)**.** A maximal subgraph of a graph in which every pair of edges lie together on some cycle.

Distinct blocks of a graph share no edges with each other, so the edges of any graph can be partitioned by which blocks they are in. Note that a block may consist of a single edge, if that edge lies in no cycles in the graph.

Definition 1.13 (Turán Number)**.** The maximum number of edges possible in a graph that does not contain a specific subgraph.

1.2. **Problem.** What is the Turán number for a k-chorded cycle? That is, what is the maximum number of edges that can be placed on a graph with n vertices such that the graph does not contain a k-chorded cycle? Throughout this paper, we attempt to show the Turán number for certain cases of k and then try to formulate a conjecture to generalize for any k.

1.3. **Previous Results.** In this section, we state several known results that will be used in the proofs of our theorems.

Theorem 1.14 (Dirac's Theorem [**Dirac**])**.** *If a graph G has n vertices and minimum degree of at least $n/2$, has a Hamiltonian cycle.*

Theorem 1.15 ([**KaraKral**])**.** *Any graph with minimum degree δ has a cycle with at least $\frac{(\delta+1)(\delta-2)}{2}$ chords.*

1.4. **Some Useful Methods.** Below we list some useful methodologies that assisted in finding patterns and developing conjectures for Turán Numbers, as well as were helpful in developing proofs.

Fact 1. *If P is a maximal length path in a graph G, and if vertex v is an end-vertex of P, then any neighbor of v in G must lie on P.*

Indeed, if v had a neighbor $w \in V(G) \setminus V(P)$, then there would be a longer path in G, obtained by appending w to the end of P. This contradicts the maximality of P.

1.5. **Graph Decomposition.** The process of graph decomposition was also used extensively in most of the theorems in this paper. Essentially, graph decomposition is the result of removing a certain amount of edges and vertices in a graph leaving a subgraph each time that satisfies certain conditions that the original graph also had. For instance, for a graph with n vertices and n edges, removing a leaf one at a time in the graph results in a subgraph with n-1 edges and n-1 vertices. Both the new subgraph and the graph have the same number of vertices and edges, but reducing the graph helps in looking at a smaller version of graphs that have hundreds or thousands of vertices and thus helps give us a better understanding of the whole graph by examining a smaller subgraph.

1.6. **Extremal Bipartite Graphs.** Extremal Bipartite graphs were useful in coming up with some of the motivation for some of the formulas in this paper. For instance, for finding the number of 4-chorded cycles in a graph, we could set up an extremal bipartite graph with 3 vertices on one side and the rest (n-3) on the other side. Connecting each of the vertices on one side to the other side gives us a total of 3n-9 edges with a graph of n vertices. Now

we can see that this gives us a 3-chorded cycle in our graph and all that is needed for our graph to have 4-chords is another edge on the right side of the graph giving us 3n-8 edges and a 4-chorded cycle. This example gives insight on how the authors of this paper came up with some of the formulas for the Turán number for some of the subcases.

2. Examples of Turán Numbers for a k-chorded cycle.

2.1. Turán Number for a Cycle.

Question 2.1. *How many edges can a graph with n vertices have without containing a cycle?*

Lemma 2.2. *A graph with minimum degree 2 must contain a cycle.*

Proof. If we start with a vertex and draw an edge to another vertex, then both vertices will have degree one. Each must have a degree of two, so we keep on drawing a new edge and a vertex to make them have at least a degree of two. This process must stop at some point because there are only a finite number of vertices and edges in the graph. The last vertex then only has a degree of one so it must share an edge with a vertex already on the path before it, creating a cycle. ■

Theorem 2.3. *If a graph has the same number of vertices and edges, then it must contain a cycle.*

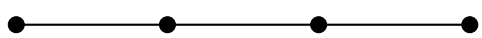
Proof. We take a graph with n vertices and n edges, and delete leaves one at a time, leaving a subgraph each time with the same number of vertices and edges. This process continues till we are left with a subgraph with no leaves and this subgraph must have at least 3 vertices and 3 edges, because it is impossible for a graph to have just 2 vertices and 2 edges, as taking a leaf out takes one vertice and edge. This means that the subgraph has a minimum degree of two, because it has no leaves. By Lemma 2.1, this subgraph must also have a cycle, which implies that the original graph must also contain a cycle. ■

Corollary 2.4. *The only connected graphs that do not have a cycle are the graphs with a minimum degree of one. In other words, every acyclic graph has minimum degree 1.*

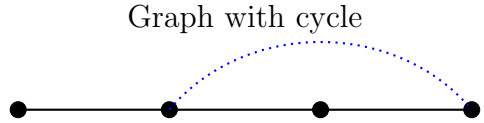
Corollary 2.5. *If a graph has a lesser number of vertices than edges, then it must contain a cycle.*

Conclusion: The Turán number of a cycle is **n-1**, because a graph with n vertices may have no cycle on n-1 edges, but as soon as you add the nth edge, it contains a cycle (by Theorem 2.3). Below we show an example - the graph consisting of black lines and vertices does not have any cycles, but as soon as we add another edge, such as the blue dotted line, a cycle forms.

Graph without cycle



Graph with cycle



2.2. Turán Number for a Chorded Cycle.

Question 2.6. *How many edges can a graph with n vertices have without containing a chorded cycle?*

Lemma 2.7. *If a graph has a minimum degree 3, then it must contain a chorded cycle.*

Proof. If we find the longest path in a graph that has a minimum degree of 3, then the last vertex of the path must share two more edges with two vertices before it on the path. If the last vertex had a neighbor not on the path, we could have found a longer path, but we already assumed that the path is the longest possible path. Then, one of these edges will form a chord inside the cycle with the other edge.

■

Theorem 2.8. *If a graph has n vertices and at least 2n-3 edges, then it must contain a chorded cycle.*

Proof. If we take a graph with n vertices and $2n-3$ edges, and then take away all the vertices that have a degree of 1 or 2, we will be left with a subgraph of either 4 vertices and 5 edges or with a subgraph with a minimum degree of 3.

Case 1: If we keep doing this process until we are left with a subgraph with 4 vertices and 5 edges, then it must contain a chorded cycle.

Case 2: If we are left with a subgraph with no more degree 1 vertices or degree 2 vertices to take out, then we must be left with a subgraph of minimum degree 3, thus, by Lemma 2, we have a chorded cycle. If we have a graph with n vertices and more than 2n-3, then the same conclusion holds, as adding edges still contains the subgraph with a chorded cycle within it.

■

Corollary 2.9. *If a graph has no chorded cycle, then it has a minimum degree of 1, 2, or 0.*

Conclusion: The Turán number for a chorded cycle is 2n-4.

2.3. **Turán Number for a 2-Chorded Cycle.**

Question 2.10. *How many edges can a graph with have without having a 2-chorded cycle?*

Note that we refer to 2-chorded cycles as doubly chorded cycles.

Lemma 2.11. *If a graph has a minimum degree of 3, then it contains a doubly chorded cycle.*

Proof. If we find the longest path in a graph that has a minimum degree of 3, then the first vertex needs to connect to at least 2 other vertices. Let the first vertex be u, and the two vertices it connects to be v_1 and v_2 respectively. Vertices v_1 and v_2 can be anywhere on the path. Now, there must be at least one vertex that is to the left of vertices v_1 and v_2, as multiple edges (having a vertex pair with 2 or more edges connecting them) are not allowed. Let this vertex to the left of v_1 and v_2 be w. Vertex w only has degree two and needs another vertex to share an edge with. So it can either share an edge with a vertex that is also to the left of v_1 and v_2, or share an edge with a vertex that is to the right of v_2. If w shares an edge with a vertex that is to the left of v_1 and v_2, then a doubly chorded cycle results with the chords being the edge from u to v_1 and the edge from w to another vertex. Instead, if w shares an edge with a vertex that is to the right of v_2, then we also have a doubly chorded cycle with the chords being the edge from w to v_1 and the edge from u to v_2. Thus, any graph with a minimum degree of 3 will result in a doubly chorded cycle. ■

Theorem 2.12. *If a graph has n vertices and at least 2n-2 edges, then it must contain a 2-chorded cycle.*

Proof. If we take a graph with n vertices and 2n-2 edges, and take away all the vertices that have a degree of 3 or less, we will be left with either a graph with minimum degree 4 or a graph that has 4 vertices and 6 edges.
Case 1: If we are left with a subgraph with 4 vertices and 6 edges, then by the previous lemma it cotains a doubly chorded cycle.
Case 2: If we are left with a subgraph with 4 vertices and 6 edges, then it contains a doubly chorded cycle. ■

Conclusion: The Turán number for a doubly chorded cycle is 2n-3.

2.4. Turán Number for a 3-Chorded Cycle.

Question 2.13. *How many edges can a graph with have without having a 3-chorded cycle?*

Conclusion: The Turán number for a 3-chorded cycle is 2n-2.

2.5. Turán Number for a 4-Chorded Cycle.

Question 2.14. *How many edges can a graph with have without having a 4-chorded cycle?*

Theorem 2.15. *If a graph has n vertices where $n \geq 7$ and at least 3n-8 edges, then it has a 4-chorded cycle.*

Proof. If we take a graph with n vertices and at least 3n-8 edges and the graph has minimum degree of 4 or more, then by Kara-Kral theorem, it has a 5-chorded cycle. If the graph has a minimum degree of 3 or less, then we decompose the graph removing low degree vertices one at a time until either the graph has minimum degree of 4 or more, in which case we use Kara-Kral's Theorem and have a 5-chorded cycle, or we are left with a graph with 7 vertices and at least 13 edges. In the later case, we have three different cases.

Case 1: If there is a degree 2 vertex v, then we remove it, leaving a subgraph G' with 6 vertices, and 11 edges. Now if G' has a vertex of degree 1 or 2, we remove it leaving 5 vertices and 9 or 10 edges. This graph is either K_5 or K_5^-, so it has a Hamiltonian cycle with either 4 or 5 chords. If G' doesn't have a vertex of degree 1 or 2, then it has minimum degree 3, so by Dirac's Theorem, it has a Hamiltonian cycle, so the other 5 edges are chords of that cycle.

Case 2: If there's a degree 1 vertex v, we remove it leaving a subgraph G' with 6 vertices and 11 edges. Now, if G' has a vertex of degree 1 or 2, we remove it leaving 5 vertices and 10 or 11 edges. Since any graph with 5 vertices can have a maximum of 10 edges without having double edges between two vertices, we remove the 5 vertices and 11 edges case. For the 5 vertices and 10 edges graph, we have a K_5 graph with 5 edges on the Hamiltonian cycle and the rest of the 5 edges the 5 chords. If G' doesn't have a vertex of degree 1 or 2, then it has a minimum degree 3, so by Dirac's Theorem it has a Hamiltonian cycle, so the other 6 edges are chords of that cycle.

Case 3: If there's not a vertex of degree 1 or 2, it has minimum degree 3. This gives us a few further subcases on the length of the longest cycle in the graph.

Subcase 1: If the longest cycle has 7 vertices, the cycle uses 7 edges, with the remaining 6 edges as the chords of the cycle.

Subcase 2: If the longest cycle has 6 vertices, then we call the 7th vertex x. If x is connected to 2 adjacent edges on the cycle, then we would make a longer cycle, contradicting our assumption, so x is only connected to at most 3 of the cycle vertices. That means we have 6 edges on the cycle, 3 edges connected to x, and the rest 4 as chords on the cycle as the 4 edges must be between the cycle vertices.

Subcase 3: If the longest cycle has 5 vertices, then there are two vertices x and y not on the cycle. As mentioned in the previous case x and y cannot be connected to adjacent vertices of the cycle, so there are at most 2 edges from x and y to the cycle each. If there is no edge between x and y, then we have 4 chords in the cycle. If there is an edge between x and y, and if both x and y have exactly 2 connections to the cycle, then we call the two vertices hat x is connected to b and d and the cycle-vertex between them c. Now if we swap the role of x and c, we now have x a cycle-vertex and c an extra vertex. We know c can only be connected to 2 cycle-vertices and those are b and d so it's not connected to any other cycle-vertices. Now the total number of edges in the graph is 5 ()for the cycle) + 2 (for the edges from c to the cycle) + 2(for the edges connected to y) +3 (for the chords of the cycle). This gives us a total of 12 edges, which is a contradiction to our previous assumption, thus this case is impossible.
For all other subcases that have the longest cycle have length of 4 or less, we can't fit 13 edges into 7 vertices, so we are done. ■

Conclusion: The Turán number for a 4-chorded cycle is 3n-9.

2.6. **Turán Number for a 5-Chorded Cycle.**

Question 2.16. *How many edges can a graph with have without having a 5-chorded cycle?*

Theorem 2.17. *If G is a graph with n vertices ($n \geq 6$) and at least 3n-7 edges, it must have a 5-chorded cycle.*

Proof. If we take a graph with n vertices and at least 3n-7 edges and remove vertices of degree 1,2, or 3, one at a time until either the graph has minimum degree of 4 or more, in which case by the Kara-Kral Theorem, our graph has a 5-chorded cycle, or our graph has 6 vertices and at least 11 edges. Now in the later scenario, we are left with two cases.

Case 1: The graph contains a Hamiltonian cycle and has 5 chords that are inside the cycle. Thus, this graph has a 5-chorded cycle.

Case 2: We are left with a graph that has the largest cycle with length 5, with the other vertex not part of the 5-cycle (x). Now x can only connect to at most two non-adjacent vertices inside the cycle. Otherwise, the graph will have a longer cycle that goes through x. If x has degree 1, then the the 5-cycle is a K_5 graph with the other 5 vertices as the chords within the cycle. If x has degree 2, then we let the two vertices that x connects to be a and c and the vertex between them be b. Now, if we swap the positions of x and b we can see that b cannot connect to any other vertex other than a and c, in order for the graph to have the largest cycle to be a 5-cycle. Then, we must satisfy the condition of our graph having

11 edges and the only way to draw that would require us to draw another edge from x or b, which is not possible. therefore, this case results in a contradiction and is impossible. ■

Conclusion: The Turán number for a 5-chorded cycle is 3n-8.

2.7. **Turán Number for a 6-Chorded Cycle.**

Question 2.18. *How many edges can a graph with have without having a 6-chorded cycle?*

Theorem 2.19. *If a graph has n vertices and 3n-6 edges, where $n \geq 7$, then it must have a 6-chorded cycle.*

Proof. If we begin with a graph with n vertices and 3n-8 edges, then if it has a minimum degree of 5 or more, then by Kara-Kral Theorem it has a 6-chorded cycle. Otherwise, we decompose the graph removing vertices of low degree one at a time until either the graph has minimum degree 5, in which case we have a 6-chorded cycle by the Kara-Kral Theorem, or we are left with a graph with 7 vertices and 15 edges. Now we have a few cases based on the length of the longest cycle in the graph.

Case 1: If the longest cycle has length 7, then the graph has a Hamiltonian cycle with 7 edges on the cycle and 8 edges as chords, giving us a 8-chorded cycle.

Case 2: If the longest cycle has length 6, then we have another non-cycle vertex ,v, on the graph which is connected to at most three non-adjacent vertices on the graph. This gives us a total of 6 edges on the cycle + 3 edges from v. Thus, the remaining 6 edges must be chords within the cycle giving us a 6-chorded cycle.

Case 3: If the longest cycle has length 5, then we have two non-cycle vertices , x and y, each of is connected to at most 2 vertices on the cycle and there may or may not be an edge between x and y. Now we have three further subcases.

Subcase 1: Both x and y send an edge to the same two vertices on the cycle. In this case, if x and y also share an edge between them, then there is a 6-cycle in the graph, which is a contradiction. Otherwise, if x and y don't share an edge, then we have 2 edges from x + 2 edges from y + 5 edges on the cycle + 5 chords inside the cycle. This gives us only 14 edges, so this case is also impossible.

Subcase 2: The vertices x and y both send one edge to exactly one common vertex on the cycle. In this case, if x and y also share an edge between them, then there is a 6-cycle in the graph, which is a contradiction. Otherwise, if x and y don't share an edge, then we have 2 edges from x + 2 edges from y + 5 edges on the cycle + 5 chords inside the cycle. This gives us only 14 edges, so this case is also impossible.

Subcase 3: The vertices x and y do not send an edge to the same vertex on the cycle. In this case, we can trace a 7-cycle on our graph giving us a contradiction of our assumption of the largest cycle having length 5. Let x be connected to the cycle-vertices a and c and let y be connected to the cycle-vertices b and d and let e be the 5th cycle vertex. In this graph, vertex b is adjacent to a and c. Then, the 7-cycle is either b-c-x-a-e-d-y-b or b-a-x-c-d-e-y-b. ■

Conclusion: The Turán number for a 6-chorded cycle is 3n-7.

3. Concluding Remarks

This paper explored the Turán Number for a k-chorded cycle up to $k = 6$. Our findings our summarized in the table below. In the table below, we summarize the Turán Number for n vertices for sufficiently large numbers (see the restrictions column).

Restrictions	Chords	Turán Number
n$\geq$ 3	0	n-1
n$\geq$ 4	1	2n-4
n$\geq$ 4	2	2n-3
n$\geq$ 5	3	2n-2
n$\geq$ 7	4	3n-9
n$\geq$ 6	5	3n-8
n$\geq$ 7	6	3n-7

Definition 3.1 (Keystone Turán Numbers)**.** Let us call the Turán Numbers where the coefficient of n and the absolute value of the constant term increase Keystone Turán Numbers.

Conjecture 3.2. *The Turán Number for a k-chorded cycle, given sufficiently large values of n, can be defined recursively as follows:*

(1) If k corresponds to a Keystone Turán Number, then the Keystone Turán Number of a k-chorded cycle is $(k+1)n - (k+1)^2$
(2) If k does not correspond to a Keystone Turán Number, then the corresponding Turán Number can be found by counting down from the previous Keystone Turán Number until reaching the current value of k.

Some topics of further work is to prove or disprove this conjecture, and apply the methods used in this paper to find Turán Numbers for different subgraphs, such as claws and H graphs.

4. Acknowledgements

We would like to thank our advisor, Dr. Bradley Elliott for guiding us through the research process. We would also like to thank the Emory Math Circle program for giving us the opportunity to undertake this project.

5. References

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